

$p(\theta|y) = \frac{p(\theta, y)}{p(y)} = \frac{p(\theta)p(y|\theta)}{p(y)}$: A **frequentist procedure** quantifies uncertainty in terms of repeating the process that generated the data **many times**. However, we express our uncertainty about θ using **probability distributions**

$X \sim \text{Bernoulli}(\theta)$ PMF: $P(X = 0) = 1 - \theta$ and $P(X = 1) = \theta$

$X \sim \text{Binomial}(N, \theta)$ PMF: $P(X = x) = \binom{N}{x} \theta^x (1 - \theta)^{N-x}$

$X \sim \text{Poisson}(N\theta)$ PMF: $P(X = x) = \frac{\exp(-N\theta)(N\theta)^x}{x!}$

$X \sim \text{Normal}(\mu, \sigma^2)$ PDF: $f(x) = \frac{1}{\sqrt{2\pi\sigma}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$

$X \sim \text{Gamma}(a, b)$ PDF: $f(x) = \frac{b^a}{\Gamma(a)} x^{a-1} \exp(-bx)$

$X \sim \text{InverseGamma}(a, b)$ PDF: $f(x) = \frac{b^a}{\Gamma(a)} x^{-a-1} \exp(-b/x)$

$X \sim \text{Beta}(a, b)$ PDF: $f(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}$

$Y|\theta \sim \text{NegativeBinomial}(n, \theta)$ $\text{Prob}(Y = y|\theta) = \binom{y+n-1}{y} \theta^y (1-\theta)^n$

The posterior mean is

$$\hat{\theta} = E(\theta|Y) = \int \theta p(\theta|Y) d\theta$$

The **posterior probability of the null hypothesis** is

$$\text{Prob}(\theta < 0.5|Y) = \int_0^{0.5} p(\theta|Y) d\theta$$

PROBLEM 1
 X_1 and X_2 have joint PMF

x_1	x_2	$\text{Prob}(X_1 = x_1, X_2 = x_2)$
0	0	0.15
1	0	0.15
2	0	0.15
0	1	0.15
1	1	0.20
2	1	0.20

PROBLEM 1
a. Marginal distribution of X_1

$$P(X_1 = 0) = \sum_{i=0}^1 P(X_1 = 0, X_2 = i) = 0.15 + 0.15 = 0.3$$

$$P(X_1 = 1) = \sum_{i=0}^1 P(X_1 = 1, X_2 = i) = 0.15 + 0.2 = 0.35$$

$$P(X_1 = 2) = \sum_{i=0}^1 P(X_1 = 2, X_2 = i) = 0.15 + 0.2 = 0.35$$

c. Conditional distribution of $X_1|X_2$

$$P(X_1 = 0|X_2 = 0) = \frac{P(X_1 = 0, X_2 = 0)}{P(X_2 = 0)} = \frac{1}{3}$$

$$P(X_1 = 1|X_2 = 0) = \frac{P(X_1 = 1, X_2 = 0)}{P(X_2 = 0)} = \frac{1}{3}$$

$$P(X_1 = 2|X_2 = 0) = \frac{P(X_1 = 2, X_2 = 0)}{P(X_2 = 0)} = \frac{1}{3}$$

$$P(X_1 = 0|X_2 = 1) = \frac{P(X_1 = 0, X_2 = 1)}{P(X_2 = 1)} = \frac{3}{11}$$

$$P(X_1 = 1|X_2 = 1) = \frac{P(X_1 = 1, X_2 = 1)}{P(X_2 = 1)} = \frac{4}{11}$$

$$P(X_1 = 2|X_2 = 1) = \frac{P(X_1 = 2, X_2 = 1)}{P(X_2 = 1)} = \frac{4}{11}$$

e. Try one value for X_1 and X_2 to show whether they are independent. For example:
 $P(X_1 = 0, X_2 = 0) \neq P(X_1 = 0)P(X_2 = 0)$, so X_1 and X_2 are not independent.

PROBLEM 1
Assume that $Y|\theta \sim \text{NegBinomial}(\theta, m)$ and $\theta \sim \text{Beta}(a, b)$.
a. Derive the posterior of θ (use Bayes' rule)

The likelihood is

$$p(Y|\theta) \propto \theta^Y (1-\theta)^m$$

The prior is

$$p(\theta) \propto \theta^{a-1} (1-\theta)^{b-1}$$

The posterior is

$$\begin{aligned} p(\theta|Y) &\propto p(Y|\theta)p(\theta) \\ &\propto [\theta^Y (1-\theta)^m][\theta^{a-1} (1-\theta)^{b-1}] \\ &\propto \theta^{Y+a-1} (1-\theta)^{m+b-1} \end{aligned}$$

Therefore, $\theta|Y \sim \text{Beta}(Y+a, m+b)$

PROBLEM 2

According to insurance.com, the 2017 auto theft rate was 135 per 10,000 residents in Raleigh NC compared to 214 per 10,000 residents in Durham/Chapel Hill. Assuming Raleigh's population is twice as large as Durham/Chapel Hill and a car has been stolen somewhere in one of these two areas, what the probability it was stolen in Raleigh?

PROBLEM 2

$P(\text{the car was stolen in R} | \text{a car was stolen})$

$$\begin{aligned} &= \frac{P(\text{a car was stolen} | \text{the car was stolen in R})P(\text{the car was stolen in R})}{P(\text{a car was stolen})} \\ &= \frac{P(\text{a car was stolen} | \text{the car was stolen in R})P(\text{the car was stolen in R})}{\sum_{i \in \{R, DCH\}} P(\text{a car was stolen} | \text{the car was stolen in } i)P(\text{the car was stolen in } i)} \\ &= \frac{(135/1000)(2/3)}{(135/1000)(2/3) + (214/1000)(1/3)} = 0.5579 \end{aligned}$$

PROBLEM 3

Your daily commute is distributed uniformly between 15 and 20 minutes if there is no convention downtown. However, conventions are scheduled for roughly 1 in 4 days, and your commute time is distributed uniformly from 15 to 30 minutes if there is a convention. Let Y be your commute time this morning.

- What is the probability that there was a convention downtown given $Y = 18$?
- What is the probability that there was a convention downtown given $Y = 28$?

PROBLEM 3

Let $C = 1$ if there is a convention and $C = 0$ otherwise.

Prior: $P(C = 1) = 1/4$

Likelihood:

$(Y|C = 0) \sim \text{Uniform}(15, 20)$, then $P(Y = y|C = 0) = 1/5$ for $15 \leq y \leq 20$

$(Y|C = 1) \sim \text{Uniform}(15, 30)$, then $P(Y = y|C = 1) = 1/15$ for $15 \leq y \leq 30$

a. Bayes' rule gives:

$$\begin{aligned} P(C = 1|Y = 18) &= \frac{P(Y = 18|C = 1)P(C = 1)}{P(Y = 18|C = 1)P(C = 1) + P(Y = 18|C = 0)P(C = 0)} \\ &= \frac{(1/15)(1/4)}{(1/15)(1/4) + (1/5)(3/4)} = 0.1 \end{aligned}$$

b. $P(C = 1|Y = 28) = 1$, since $Y = 28$ can only be possible when there is a convention. (you can also use Bayes' rule).

PROBLEM 2

In a study that uses Bayesian methods to forecast the number of species that will be discovered in future years, a study reports that the number of marine bivalve species discovered in 2015 was 20. Denoting Y as the number of species discovered and assuming $Y|\lambda \sim \text{Poisson}(\lambda)$ and $\lambda \sim \text{Uniform}(0, 100)$

a. Derive the posterior distribution of λ (use Bayes' rule)

PROBLEM 2

a. Model: $Y|\lambda \sim \text{Poisson}(N\lambda)$

$\lambda \sim \text{Uniform}(0, 100)$

The likelihood is:

$$\begin{aligned} p(Y|\lambda) &\propto \exp(-N\lambda)(N\lambda)^Y \\ &\propto \exp(-N\lambda)\lambda^Y \\ &= \exp(-\lambda)\lambda^Y \text{ since } N = 1. \end{aligned}$$

The prior is:

$$p(\lambda) = \frac{1}{100}$$

The posterior is:

$$\begin{aligned} p(\lambda|Y) &\propto p(Y|\lambda)p(\lambda) \\ &\propto (\exp(-\lambda)\lambda^Y) \frac{1}{100} \\ &\propto \exp(-\lambda)\lambda^Y \\ &= \lambda^{Y+1-1} \exp(-\lambda) \end{aligned}$$

Therefore, $\lambda|Y \sim \text{Gamma}(Y+1, 1)$

PROBLEM 3

The table below has the overall free throw proportion and results of free throws taken in pressure situation, defined as “clutch” (<https://stats.nba.com/>) , for five National Basketball Association players for the 2016 – 2017 season.

Player	Overall proportion	Clutch makes	Clutch attempts
Stephen Curry	0.898	24	26
Giannis Antetokounmpo	0.770	28	41
John Wall	0.801	66	82
LeBron James	0.674	27	39
Kevin Durant	0.875	13	16

The parameter of interest is the clutch success probability $\theta \sim Uniform(0,1)$. Denote N the number of clutch attempts and Y the number of clutch makes. We assume that each attempt is independent of each other and in each attempt the probability of success is θ . Therefore, the likelihood is $Y|N \sim Binomial(N, \theta)$.

- a. Derive the posterior distribution of θ (use Bayes’ rule)
- b. Plot the posteriors of the clutch success probabilities (for five players in one plot)
- c. Conduct hypothesis testing to confirm if any of the players have a different clutch percentage than overall percentage (one hypothesis testing for each player)

c. Define θ_0 as the overall percentage. Then, the hypothesis is

$H_0: \theta = \theta_0$
 $H_1: \theta \neq \theta_0$

or

$H_0: \theta \leq \theta_0$
 $H_1: \theta > \theta_0$

Note: Ideally, we use the second hypothesis statement since $P(\theta = \theta_0)$ will always be 0. I solve the problem using the second hypothesis. But, due to the wording in the question, using the first hypothesis is also correct.

The posterior probability of H_0 is written as

$$P(\theta \leq \theta_0 | Y) = \int_0^{\theta_0} p(\theta | Y) d\theta = \int_0^{\theta_0} \frac{\Gamma(N + 2)}{\Gamma(Y + 1)\Gamma(N - Y + 1)} \theta^Y (1 - \theta)^{N - Y} d\theta$$
$$= \frac{\Gamma(N + 2)}{\Gamma(Y + 1)\Gamma(N - Y + 1)} \int_0^{\theta_0} \theta^Y (1 - \theta)^{N - Y} d\theta$$

However, we don’t need to solve the integral above since we can use Monte Carlo sampling to compute $P(\theta \leq \theta_0 | Y)$ for each player. We reject $H_0: \theta \leq \theta_0$ if $P(\theta \leq \theta_0 | Y)$ is small. We can also conclude if $P(\theta \leq \theta_0 | Y)$ is close to 0 or 1, then the player has a different clutch percentage than overall percentage.

The calculation and conclusion for each player is presented below:

Player	$P(\theta \leq \theta_0 Y)$	Conclusion
SC	0.471	Not different
GA	0.920	Different
JW	0.507	Not different
LJ	0.437	Not different
KD	0.846	Different

PROBLEM 3

Define N as the number of clutch attempts and Y as the number of clutch made.

- a. Model: $Y | \theta \sim Binomial(N, \theta)$
 $\theta \sim Uniform(0,1)$

The likelihood is

$p(Y | \theta) \propto \theta^Y (1 - \theta)^{N - Y}$

The prior is

$p(\theta) = 1$

The posterior is

$$p(\theta | Y) \propto p(Y | \theta)p(\theta)$$
$$\propto \theta^Y (1 - \theta)^{N - Y}$$
$$= \theta^{Y + 1 - 1} (1 - \theta)^{N - Y + 1 - 1}$$

Therefore, $\theta | Y \sim Beta(Y + 1, N - Y + 1)$